

Installing Kicad:

We will use a docker container image of kicad. Our container environment „apptainer“ can read the docker container format:

```
apptainer run docker://kicad/kicad:nightly
```

In the zitipool, we use a precompiled image:

```
apptainer run /shares/zitipoolhome/nca-opt/kicad_image.sif
```

The first time it will download the docker container. If you use it a second time, it will use the cached image. If the docker images changes on the docker server, apptainer will automatically update it.

In the apptainer environment, we start kicad (and detach it from the shell, since it is a GUI application):

```
apptainer> kicad &
```

In the kicad window, select „new projcet“.

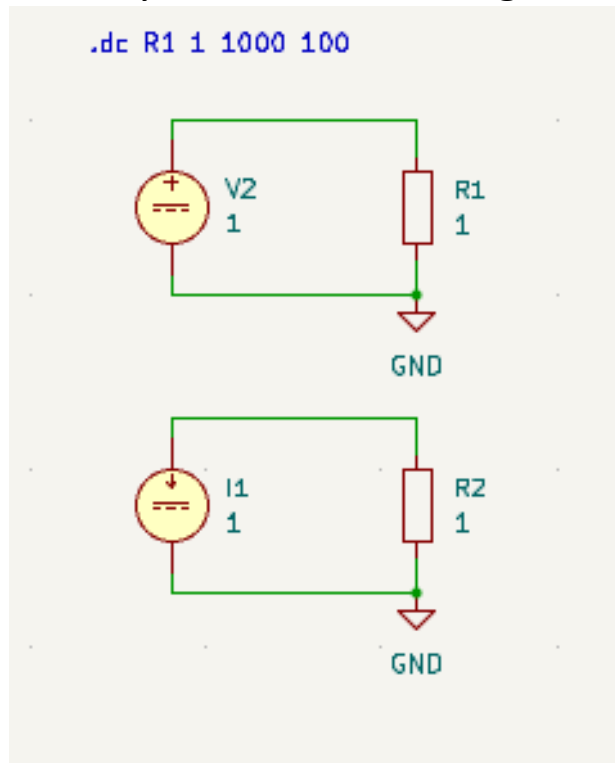
Lesson 1: basic circuits

The brain uses electricity to communicate between nerve cells. In neuromorphic computing we want to emulate the brain with electronic circuits. The following lesson will introduce the necessary basic concepts.

We always start from a closed electrical circuit, connecting at least one energy source to a number of resistors and capacitors.

KiCad lesson 1 (Kc1)

Create two elementary circuits, one using a voltage source and one a current source:



To understand voltage and current at the resistors, we use Ohm's law: $R = U/I$

Lesson 1: basic circuits – current and voltage sources – ideal sources

Q1.1:

Run the simulation and observe voltage and current at the resistors for different values of V, I and R1 and R2. You should be able to verify Ohm's law for the resistors. Write it down:

Hint: select DC Sweep Analysis and sweep over V2 and I1: plot voltage vs current for both circuits

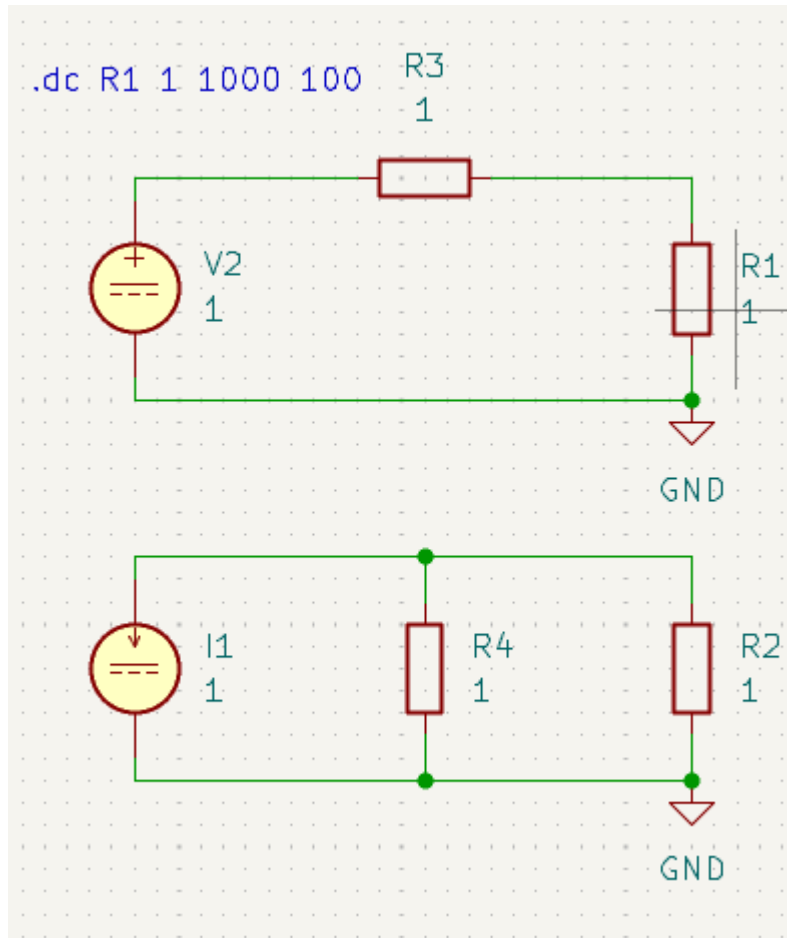
Q1.2:

Read out the values of V, I for the two sources while varying R1 and R2, respectively. How does it change? Can you deduce the equivalent resistances for the two sources?

Hint: select DC Sweep Analysis and sweep over R1 and R2: plot V(R1) and I(R2) vs V2 and I1 respectively

Lesson 1: basic circuits – current and voltage sources – real sources

KC2: Extend the schematic to include a resistor resembling the internal resistance of the source. The combination of an ideal source with a resistance is called a real source:



Lesson 1: basic circuits – Kirchhoff's laws – calculating voltages and currents of a circuit

If we add more components to the circuit, we need some rules to derive the equations for all voltages and currents in the circuit -> Kirchhoff's and Ohm's laws (including ideal sources)

Use them to solve the next question:

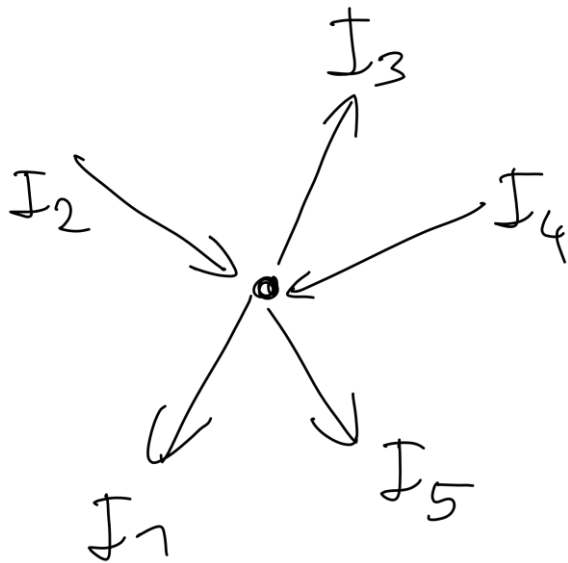
Q2.1: What is the VDC to IDC ratio that always produces the same voltage at R1 resp. R2 for a given R_s ? Is it possible to deduce the nature of the source if you only know V and I?

Lesson 1: basic circuits – Kirchhoff's laws - KCL

Gustav Robert Kirchhoff, 1824-1887, was a professor in Heidelberg from 1854 to 1876, the KIP Logo is a reference to his work

Kirchhoff's Current Law: KCL

The sum of all currents entering and leaving any node of a circuit is always zero.



Convention: currents leaving the node are positive

$$I_1 - I_2 + I_3 - I_4 + I_5 = 0$$

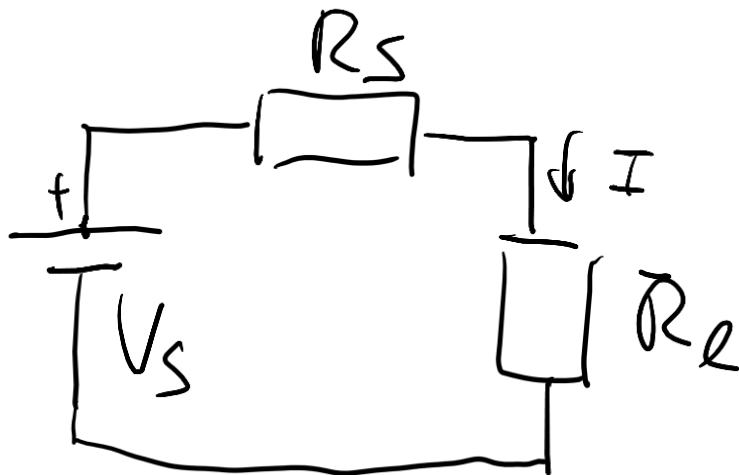
Underlying law of physics: charge conservation

Kirchhoff's Voltage Law, KVL

The sum of all voltages along any closed loop in an electrical circuit is zero.

Example:

$$-V_s + I \cdot R_s + I \cdot R_l = 0$$



Conventions:

1. Positive voltage drop along the direction of current flow in a resistor
2. Negative voltage drop along a voltage source from – to +

Note: KCL is used to argue that the current I is the same everywhere in the circuit.

Underlying law of physics: conservation of energy

What does „voltage drop“ mean? **Voltage = energy per charge**

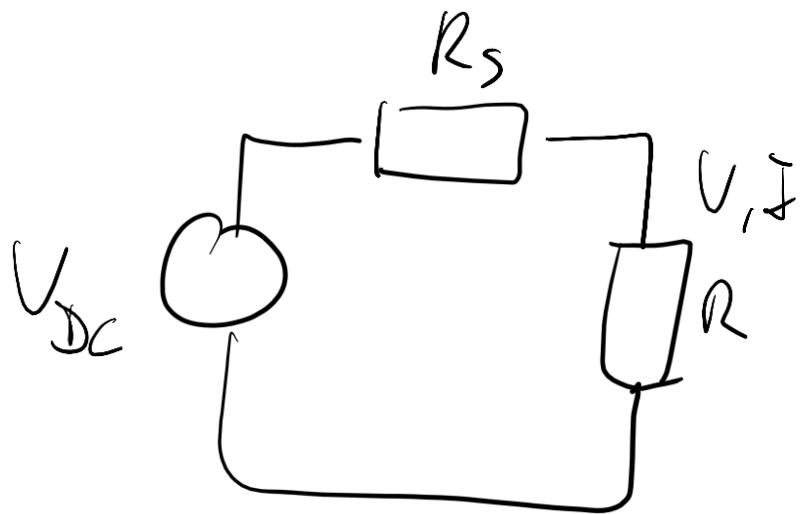
Voltage drop (Volt):

Energy (Joule) the charge (Coulomb) loses by passing the resistor.

Where does the energy go -> heat!

Thus, to keep a constant current (charge per time) flowing in a circuit loop, no energy must be lost. All losses have to be compensated by energy sources (i.e. batteries) **→ Sum of all voltages = 0**

Hint to Q2.1



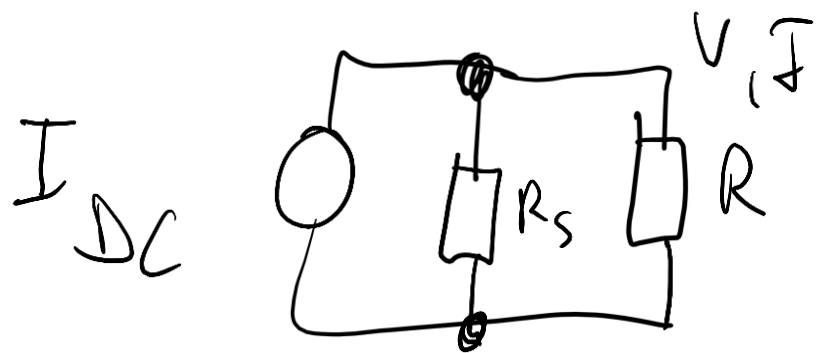
$$V = V_{DC} - R_S \cdot I \quad (1) \quad \text{KVL}$$

$$I = I_{DC} - \frac{V}{R_S} \quad (2) \quad \text{KCL}$$

(1) in (2):

$$I = I_{DC} - \frac{V_{DC}}{R_S} + I$$

$$\frac{V_{DC}}{I_{DC}} = R_S$$



Lesson 1: basic circuits – capacitance

Time plays an important role in the brain. The circuits we have looked at so far are not dependent on time. Time dependency enters the picture if we allow the storage of charge (-> cell membrane). The electrical component that allows us to store charge is the capacitor:

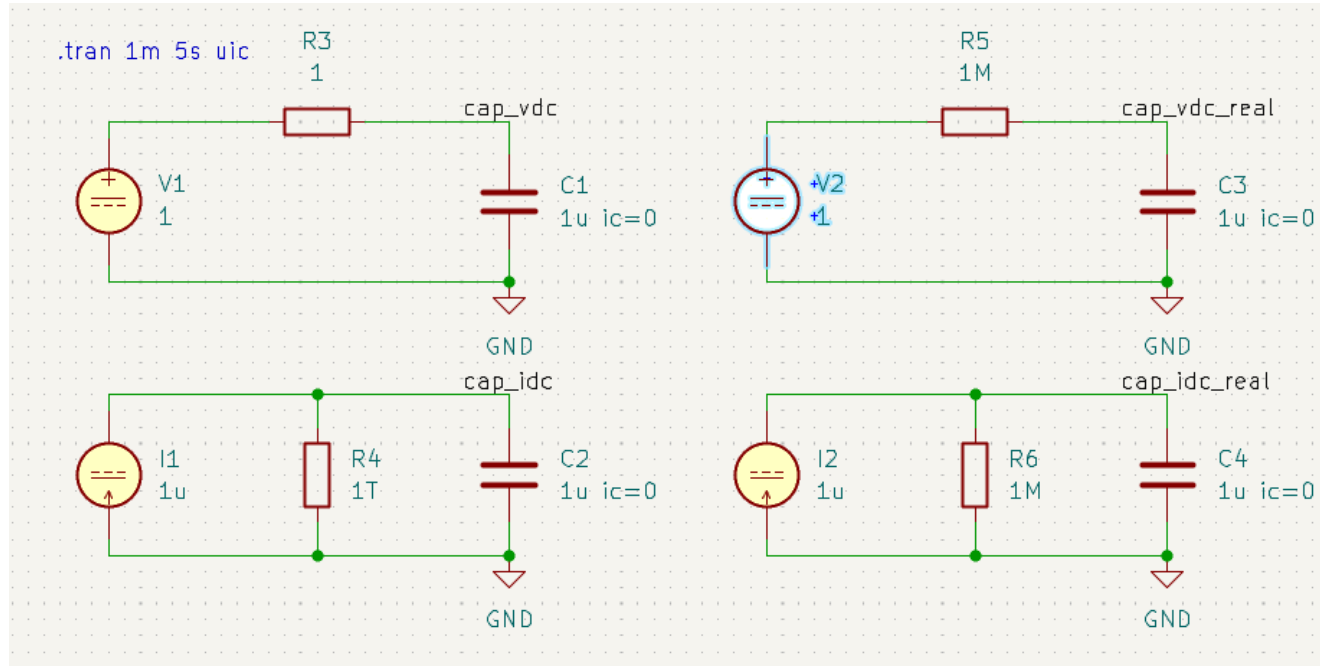
$$V_{\text{capacitor}} = Q_{\text{stored}} / \text{capacitance} \quad \text{capacitance } C[\text{Farad}] = \text{const}$$

Q_{stored} is the sum of all charge that has flown on the capacitor from $t=-\infty$ to now. With $I(t)=dQ(t)/dt$ one gets:

$$Q = \text{Integral}(I(t)dt) \rightarrow V(t) = \text{Int}(I(t)dt) / C$$

Lesson 1: basic circuits – capacitance – charging and discharging curves

KC3: replace the resistors R1 and R2 by capacitors.



For the simulation to operate properly we have to specify the charge on the capacitor at the beginning of our simulation (we do not integrate from $t=-\infty$, but some arbitrary time $t=0$). We set this charge to zero by adding „ $ic=0$ “ behind the value of the capacitance (just separated by space). Ic is short for „initial condition“.

To simulate the charging and discharging of capacitances, we have to simulate over time: select „transient analysis“ in the simulator and check the „use initial conditions“ box.

Lesson 1: basic circuits – capacitance – charging and discharging curves

Q3.1: What mathematical function describes $V(t)$ if the capacitor is connected to an ideal current source?

Hint: look at $V_{\text{cap_idc}}(t)$ in the lower left schematic

Q3.2: Compare $V_{\text{cap_idc_real}}(t)$ with $V_{\text{cap_vdc_real}}(t)$ to inspect the case of a real source

Hint: use the values of $R5$ and $R6$ given in the schematic first, then vary $R5$ and $R6$. Compare with the result from Q2.1 in both cases.